

Modeling Flash Dynamics in Biofilms under flow

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0.1. Introduction

The flash first described quantitatively by Harvey [1], has since been investigated by Johnson, van Schouwenberg, and van der Burg [2] and Chance, Harvey, Johnson, and Xillikan [3]. Despite significant progress in understanding these phenomena, no mathematical model currently explains the specific periphery-to-center oscillations observed in biofilms, a novel discovery by Nuno M. Oliveira not previously addressed in the literature. Therefore, this study aims to model the temporal dynamics of these flashes by focusing on the interactions between oxygen and luciferin, two key components in the bioluminescence process. The primary objective of this research is to mathematically model the flashes currently in the literature by incorporating competitive substrate kinetics, thereby providing deeper insight into the underlying mechanisms driving the temporal dynamics of bioluminescent flashes in biofilms. Bioluminescence, is a chemical reaction generally pre-dominant in deep marine species but also a phenomena of intriguing research in fireflies [4, 5]. It normally involves a Enzyme Luceferase and a substrate luciferin, Luceferase reacts with luciferin in the presence of oxygen , to produce oxyluciferin and a photon of light [6], the intensity of the light varies with each organism as we will see hastings experiments are with fungi and bacteria as in comparison to Nuno M. Oliveira experiments with Biofilms underflow. Biofilms are a 'structured community of bacterial cells enclosed in a self-produced polymeric matrix and adherent to an inert or living surface.' [7]. As to the intuition of the flash one can refer to the twitch observed from firefly light, which intuitively is observed in bacteria and biofilms. one being the system in Hastings and the other authors menioned above where the species *Vibrio fischeri* are being used pre-dominantly in suspension and the other being the experiments of Nuno M. Oliveira, with bacterial biofilms being surface attached.

Nomenclature

$[L_0]$ Initial concentration of luciferin

$[L_r]$ Reduced luciferin concentration, represented as Y

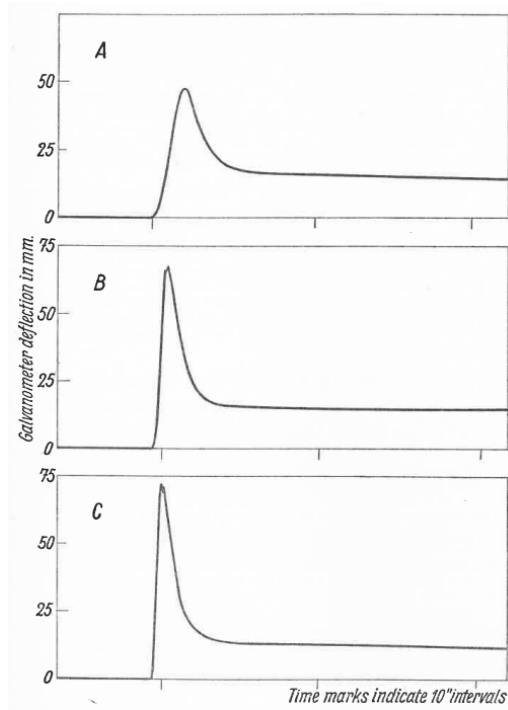
$[L_{\text{oxi}}]$ Oxyluciferin concentration, represented as $[L_0] - [L_r] = Y_0 - Y$

$[O_2]$ Oxygen concentration, represented as X

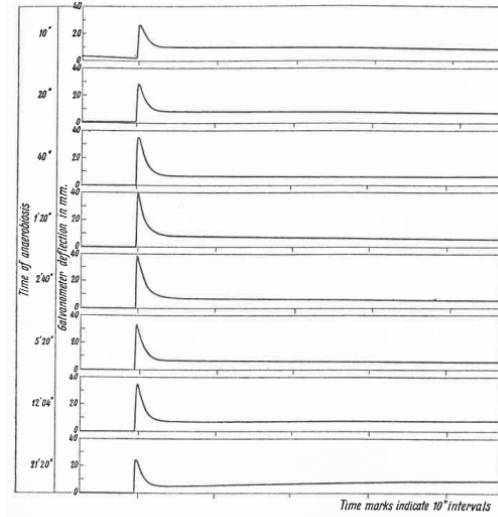
X Variable representing $[O_2]$

X_0 Initial concentration of $[O_2]$

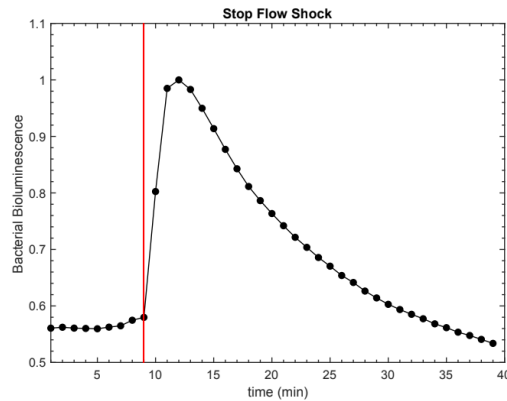
Y Variable representing $[L_r]$



(a)



(b)



(c) Plot from experiments of Nuno M. Oliveira

Figure 1. The first two subfigures (a, b) use a logarithmic time scale (times two scale), while the third subfigure (c) uses a linear time scale (minute scale). There is a notable difference between the experiments: in the experiments by Johnson and Hastings, the curve decays from the maximum and approaches a constant luminous intensity as O_2 is introduced at a constant rate. In contrast, in Nuno's experiment, the intensity decays to zero, indicating no remaining luminous intensity in the system.

0.2. Systems A and B

To conceptualize the experiments, consider two systems: System A, a closed box, and System B, a box with an attached oxygen inlet. The inlet of O_2 being closed results in both systems being closed systems, with no influx of O_2 from the external environment. This is characterised as $[O_2]_{t=0}$ at later times a the system approaches an anaerobic state at $[O_2]_{t=a}$ and if the anaerobic state is allowed to subside for time x at concentra-

tion $[O_2]_{t=a+x}$ - though the intensity of the flash is dependant on the duration allowed to subside for anaerobias as Johnson discusses [2] - and the inlet of O_2 is opened allowing a constant influx of O_2 at a steady rate, a off-shoot intensity of light -anomalous to normative intensities- has been observed[8]. Shockingly, a flash is observed in system A aswell, but without a "re-introductory effect", or as Nuno hypothesises the "re-introductory effect is from within the cell" considering the competition between respiration and bioluminescent, namely under anaerobias the respiratory tract is blocked and the bioluminescent reaction scavenges the O_2 in the system without a competitor, a loss of competition for the bioluminescent reaction plays the role of a re-"introductory effect", due to the boost in O_2 though minimal. However System A is with bacterial Biofilms and the latter with fungi, Luciferin from one group of luminous animals will not react with the luciferase from another, and vice versa [6]. Luciferin is an entirely different organic compound in different species, therefore it is expected that the mechanisms of light production are different in different organisms [9].



Figure 2. Left box = System A and Right box = System B as in [8]

0.3. Chemical Reaction Contributions

To reduce the system we consider negligibility of other extracellular reactions that are consuming or forming both oxygen and luciferin. The reason why we track the change in Luciferin and Oxygen is due to their competition and also them being the basis of our hypothesis, please see ??.

0.3.1. The Bioluminescent Reaction

The bioluminescence reaction involves the enzyme luciferase and the substrate luciferin reacting in the presence of oxygen to produce oxyluciferin and light (hv) [10, 11]. Our reaction scheme follows a two-substrate, one-enzyme system. Considering the competitive adsorption of each substrate (oxygen and luciferin) as suggested by Harvey in 1952 [8], we adopt the following expressions:

$$\frac{\frac{V}{K_m^n} [O_2]^n}{1 + \frac{[O_2]^n}{K_m^n} + \frac{[L_r]^n}{K_m^n}}$$

and

$$\frac{\frac{V'}{K_m'^n} [L_r]^n}{1 + \frac{[O_2]^n}{K_m^n} + \frac{[L_r]^n}{K_m'^n}}$$

where the Hill constant $n \in \mathbb{R}$, and:

- $V = k_2 E_0$ and $V' = k'_2 E_0$ are the limiting rates.
- $K_m = \frac{k_{-1} + k_2}{k_1}$ and $K'_m = \frac{k'_{-1} + k'_2}{k'_1}$ are the Michaelis constants for the two reactions.

For each substrate's consumption in the bioluminescent reaction, the scheme is chosen as each substrate exhibits Michaelis-Menten kinetics when analyzed independently [12].

0.3.2. Reduction of Oxyluciferin

Oxyluciferin is recycled back to reduced luciferin, allowing the cycle to repeat itself [13]. This reduction reaction occurs continuously at all times t , under both aerobic and anaerobic conditions, resulting in a system of "auto-generated light production." Assuming simple mass action kinetic for the reactions in 0.7, the following term is derived; $\frac{d[L_r]}{dt} = k_a[L_{oxi}] = k_a([L]_0 - [L]_{red})$ where the rate constant $k_a \in \mathbb{R} > 0$. We assume the total enzyme concentration E_0 to be constant in our system. Additionally, to simplify the model for the flash, we assume that the total concentration of luciferin in the system remains unchanged [?]: $[L]_0 = [L]_{red} + [L]_{oxi}$

0.3.3. Dark Reaction

Furthermore, it has been shown that oxygen can oxidize luciferin in the absence of luciferase under aerobic conditions [14], a process referred to as "dark oxidation." In this process, molecular oxygen combines with reduced luciferin to produce oxyluciferin, affecting both the rates for oxygen and reduced luciferin: $\frac{d[O_2]}{dt} = -k_d[O_2][L_r]$ and $\frac{d[L_r]}{dt} = -k_d[O_2][L_r]$ where the rate constant $k_d \in \mathbb{R} > 0$. However, it is worth considering whether k_d is different for Luciferin and oxygen respectively.

0.3.4. Respiration

On the basis of oxygen consumption being independent of oxygen pressure for 21 percent to 3 percent [15], also factors affecting Luminescence also affect respiration though no proportionality between the two is recorded, this has sufficed for these processes to considered intertwined [16], this is a significant measure in our system where the function for respiration follows Michael Menton [17] :

$\frac{d[O_2]}{dt} = -\frac{k_r[O_2]}{k_r + [O_2]} = -\frac{k_r[O_2]}{R}$ where we let $k_r + [O_2] = R$, and the respective rate constant and michaelis constant k_r and $k_r \in \mathbb{R} > 0$.

This term quantifies the rate at which oxygen is consumed in a respiratory process. As the oxygen concentration $[O_2]$ increases, the rate of oxygen consumption approaches the maximum rate characterized by k_r . When $[O_2]$ is low, the rate of consumption is limited by the availability of oxygen and approaches zero. The half-saturation constant k_r provides a measure of the sensitivity of the system to changes in oxygen concentration.

0.4. Hypothesis

During aerobic conditions when the oxygen pipe is open in our bacteria cell suspension, oxygen is diffusing into the bacteria and oxygen which is being consumed, we assume that the diffusion of oxygen in our bacterial suspension and consumption is at a constant rate in $mol/cells^{-1}$ under anaerobic conditions. Laslty we focus on the luciferin in the

system, this is $\beta([L_r]_0 - [L_r]_t)$ where we have the total luciferin in the cell and the luciferin at later times t being consumed in the system, though we only consider it being oxidised and there's no loss of total luciferin in the system as previously stated. The flux is more correctly described as the following: The *oxygen* diffusing in the bacteria - The *oxygen* being consumed (or used up in the bacteria) a constant

0.4.1. Model Equations

Quantity Name	Typical Symbol	Name of Unit	Unit Symbol
Time	t	Second	s
Amount of Substance	n	Mole	mol
Luminous Intensity	I_v	Candela	cd
Concentration	c	Moles per cubic meter	mol m ⁻³

Table 1. Units and Their Symbols

Combining the terms from sec "chemical contributions" with there respective constituents.

0.4.2. Scheme 2 (Langmuir Functional form)

$$\frac{dX}{dt} = F - k_dXY - \frac{X}{X + K_r} - K \frac{\frac{X^n}{K_m^n} \frac{Y^n}{K_m'^n}}{(1 + \frac{X^n}{K_m^n} + \frac{Y^n}{K_m'^n})^2} \quad (1)$$

$$\frac{dY}{dt} = k_a(Y_0 - Y) - k_dXY - K \frac{\frac{X^n}{K_m^n} \frac{Y^n}{K_m'^n}}{(1 + \frac{X^n}{K_m^n} + \frac{Y^n}{K_m'^n})^2} \quad (2)$$

0.4.3. Parameter Sensitivity

0.4.4. Scheme 1

$$\frac{dX}{dt} = k_c(X_0 - X) - k_dXY - \frac{k_r X}{X + K_r} - \frac{\frac{V}{K_m^n} X^n}{1 + \frac{X^n}{K_m^n} + \frac{Y^n}{K_m'^n}} \quad (3)$$

$$\frac{dY}{dt} = k_a(Y_0 - Y) - k_dXY - \frac{\frac{V'}{K_m'^n} Y^n}{1 + \frac{X^n}{K_m^n} + \frac{Y^n}{K_m'^n}} \quad (4)$$

Scaling the parameters

Given the substitutions:

Given the substitutions:

$$X = \hat{X}x_c$$

$$Y = \hat{Y}y_c$$

$$t = \hat{t}\tau$$

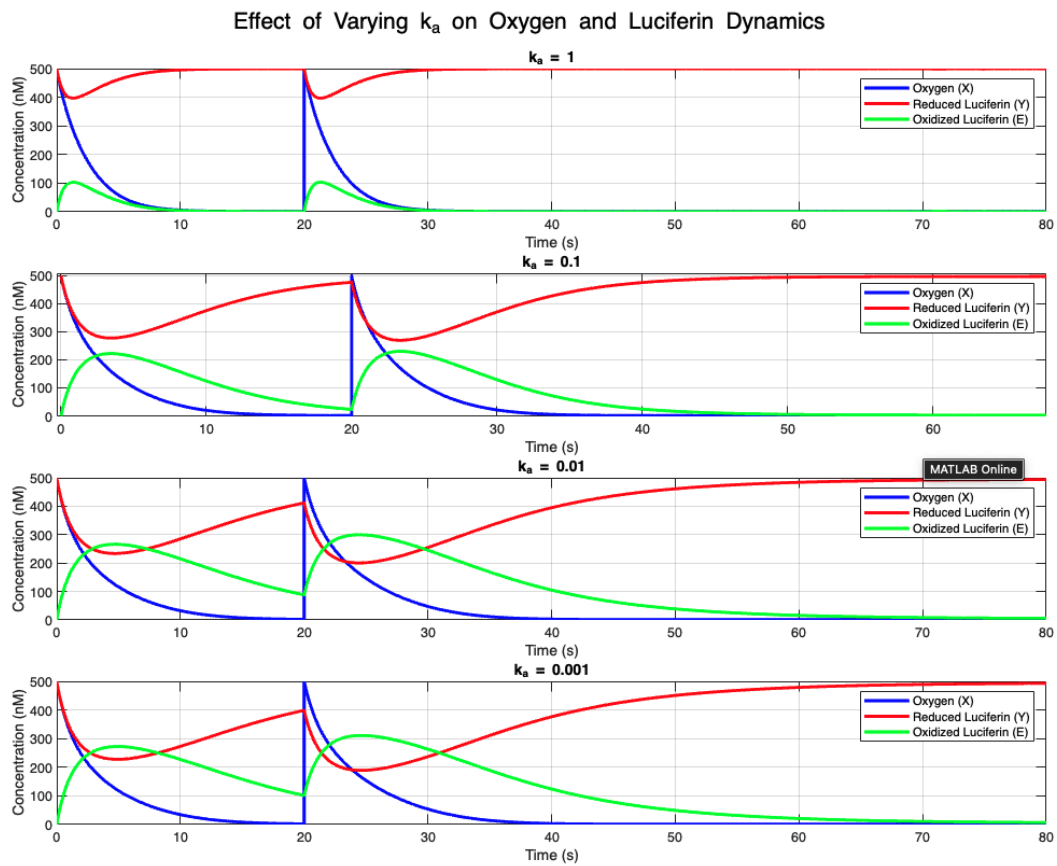


Figure 3. The plot above shows a range of k_a values, and their affect on the plots. k_a being the rate at which oxyluciferin is reduced back to luceferin namely the rate at which oxyluciferin is converted to reduced luciferin

The derivatives will transform as:

$$\frac{dX}{dt} = \frac{d(\hat{X}x_c)}{d(\hat{t}\tau)} = \frac{x_c}{\tau} \frac{d\hat{X}}{d\hat{t}}$$

$$\frac{dY}{dt} = \frac{d(\hat{Y}y_c)}{d(\hat{t}\tau)} = \frac{y_c}{\tau} \frac{d\hat{Y}}{d\hat{t}}$$

Given the substitutions:

$$\tau = \frac{1}{k_p + k_a}, \quad x_c = X_0 = 2K_m, \quad y_c = Y_0 = 2K'_m$$

0.4.5. The choice for $\hat{X}_0$ and $\hat{Y}_0$ in terms of their respective michaelis constant

The Michaelis constant represents the concentration of the substrate when the reaction reaches half of its maximal rate, this is when the enzyme reaches half saturation by the substrates on their corresponding active-sites.

It is apostulated that the actual substrate concentration is much higher it's respective Michaelis constant. We approximate that the $[S] \geq 10 \times K_m$, where K_m is the michaelis constant for $[S]$, then $v \approx v_{max}$

Therefore at this concentration the enzyme is almost full saturated, leading to the maximal rate. after this point, if we have an excess of the substrate it has minimal affect on reaction rates.

This argument shows that we can express $[S]$ in terms of K_m namely:

$$[S] = nK_m, \text{ where } n \in \mathbb{R} \text{ and } n \geq 10$$

0.4.6. n being timescale dependant

The transformed equations become:

$$\frac{d\hat{X}}{d\hat{t}} = \alpha_4 (1 - \hat{X}) - \alpha_5 \hat{X} \hat{Y} - \alpha_6 \frac{\hat{X}}{2\hat{X} + \alpha_7} - \alpha_8 \frac{(2\hat{X})^n}{1 + (2\hat{X})^n + (2\hat{Y})^n} \quad (5)$$

$$\frac{d\hat{Y}}{d\hat{t}} = (1 - \hat{Y}) - \alpha_2 \hat{X} \hat{Y} - \alpha_3 \frac{(2\hat{Y})^n}{1 + (2\hat{X})^n + (2\hat{Y})^n} \quad (6)$$

Where we define:

$$\alpha_1 = k_p + k_a, \quad \alpha_2 = 2K_m \frac{k_d}{\alpha_1}, \quad \alpha_3 = \frac{V'}{2K'_m \alpha_1},$$

$$\alpha_4 = \frac{k_c}{\alpha_1}, \quad \alpha_5 = \frac{2k_d K'_m}{\alpha_1}, \quad \alpha_6 = \frac{k_r}{K_m \alpha_1}, \quad \alpha_7 = \frac{K_r}{K_m}, \quad \alpha_8 = \frac{V}{2K_m \alpha_1}$$

It is noted that all parameters defined are unitless except α_2 and α_5 , which are in nano Molars, therefore we replace them by new parameters:

$$\gamma = \frac{\alpha_2}{K'_m}, \beta = \frac{\alpha_5}{K_m}$$

Thus,

$$\frac{d\hat{X}}{d\hat{t}} = \alpha_4 (1 - \hat{X}) - \beta \hat{X} \hat{Y} - \alpha_6 \frac{\hat{X}}{2\hat{X} + \alpha_7} - \alpha_8 \frac{(2\hat{X})^n}{1 + (2\hat{X})^n + (2\hat{Y})^n} \quad (7)$$

$$\frac{d\hat{Y}}{d\hat{t}} = (1 - \hat{Y}) - \gamma \hat{X} \hat{Y} - \alpha_3 \frac{(2\hat{Y})^n}{1 + (2\hat{X})^n + (2\hat{Y})^n} \quad (8)$$

0.4.7. Analysing the system as $\hat{X} \rightarrow 0$

eq (5) reduces to:

$$\frac{d\hat{X}}{d\hat{t}} = \alpha_4 \quad (9)$$

Solving for $\hat{X}$ and evaluating at initial conditions we get:

$$\hat{X}(t) = \alpha_4 t + \hat{X}_0 \quad (10)$$

Stability Analysis and Simplification

The simplified differential equations are:

$$\frac{d\hat{X}}{d\hat{t}} = (1 - \hat{X}) - \alpha_2 \hat{X} \hat{Y} - \alpha_3 \frac{\hat{X}}{\hat{X} + \alpha_5} - \gamma \alpha_2 \frac{\hat{X}^n}{1 + \hat{X}^n + \hat{Y}^n} \quad (11)$$

$$\frac{d\hat{Y}}{d\hat{t}} = \alpha_1 (1 - \hat{Y}) - \beta \alpha_2 \hat{X} \hat{Y} - \gamma' \alpha_2 \frac{\hat{Y}^n}{1 + \hat{X}^n + \hat{Y}^n} \quad (12)$$

Parameter Definitions

The parameters are defined as follows:

$$\alpha_1 = \frac{k_p + k_a}{k_c}, \quad \alpha_2 = K_m \frac{k_d}{k_c}$$

$$\alpha_3 = \frac{k_r}{K_m k_c}, \quad \alpha_5 = \frac{K_r}{K_m}, \quad \beta = \frac{K'_m}{K_m}$$

$$\gamma = \frac{V}{K_m \alpha_2}, \quad \gamma' = \frac{V'}{K'_m \alpha_2}$$

0.5. Results and Discussion

Total Enzyme is a conserved quantity, meaning $dE/dt + dC/dt = 0$, and therefore $[E] + [C] = [E_0]$

0.6. Conclusions

0.7. Nomenclature and Chemical equations

Reduction reaction



Dark oxidation



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